

Experiment 2: GPIO-Based Environmental Sensor Node with Real-Time Threshold Alerts

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1. Project Objective & Zero-Latency Architecture

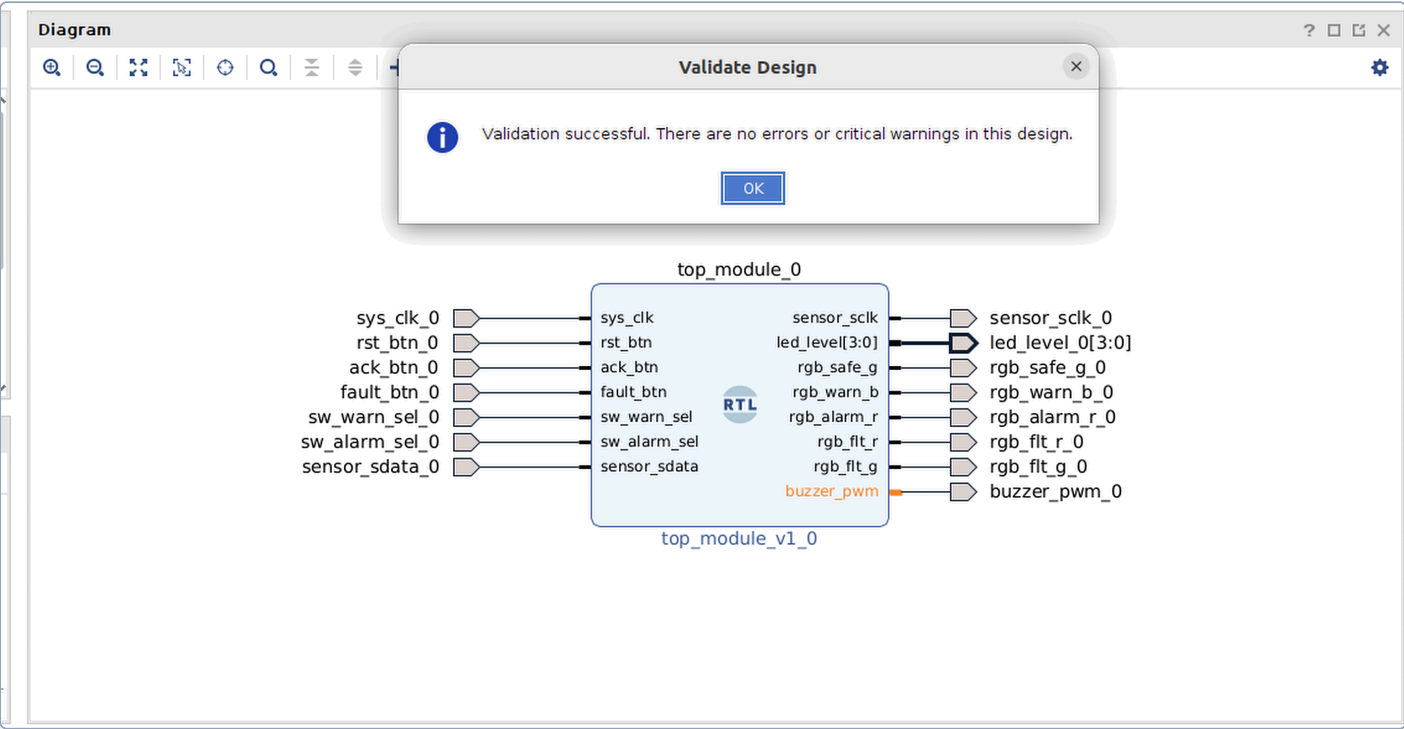
Critical industrial environments demand guaranteed sub-microsecond responses to toxic gas accumulation, flammable vapors, or thermal spikes. Conventional software microcontrollers suffer from polling overhead and scheduling jitter. This project implements an entirely hardware-deterministic sensor node on the AMD PYNQ-Z2 FPGA fabric, integrating serial digital frame sampling, on-chip moving average noise attenuation, multi-tier threshold evaluation, and dual-tone PWM acoustic signaling.

1.1 Subsystem Specifications & Hardware Interfaces

Subsystem	Verilog Module	FPGA Pin	Hardware Description
Clock Divider	clock_divider.v	H16	125 MHz scaled to 1 kHz sampling tick, 1 Hz tick, 2 Hz alert blink
Serial Sampler	sensor_sampler.v	Y18, Y19	Serial Data In (Pmod JA Pin 1), Serial Clock Out (Pmod JA Pin 2)
Digital Noise Filter	digital_filter.v	Internal RTL	4-tap FIR moving average accumulator (sum >> 2), 0 DSP slices
Threshold Engine	threshold_comparator.v	M20, M19	Multi-tier Safe / Warning / Critical Alarm classification with hysteresis
Dual-Tone Buzzer	pwm_alarm_buzzer.v	Y16	950 Hz warning chirp / 3.8 kHz siren on Pmod JA Pin 3
Status LEDs	top_module.v	R14-M14, LD4, LD5	Nibble value LEDs (LD0-LD3), RGB LD4 (Alert), RGB LD5 (Fault)
User Controls	debouncer.v	D19, D20, L19	BTN0 (Alarm Ack/Snooze), BTN1 (Synthetic Fault), BTN3 (Reset)

Figure 1: Vivado IP Integrator Top-Level Block Design

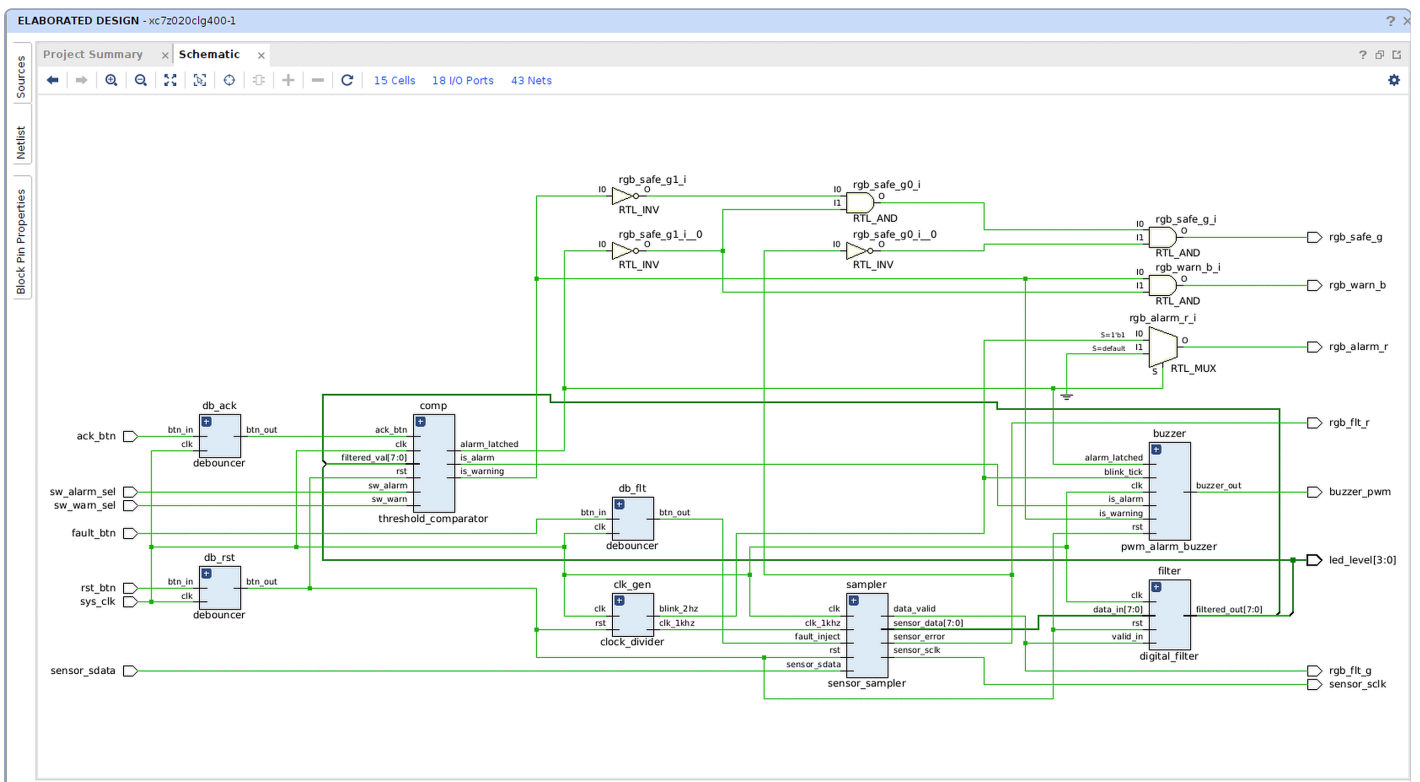
AMD PYNQ-Z2 SoC IP Block Diagram — Status: Validated (0 Errors, 0 Warnings)



Practical Vivado Verification: The IP block diagram illustrates top_module_0 mapped to system clock sys_clk_0, reset button rst_btn_0, acknowledge button ack_btn_0, synthetic fault button fault_btn_0, threshold selection switches, serial sensor interface, status LEDs, and PWM buzzer out. Design rule validation completed with 0 errors and 0 warnings.

Figure 2: Vivado Elaborated & Synthesized RTL Schematic

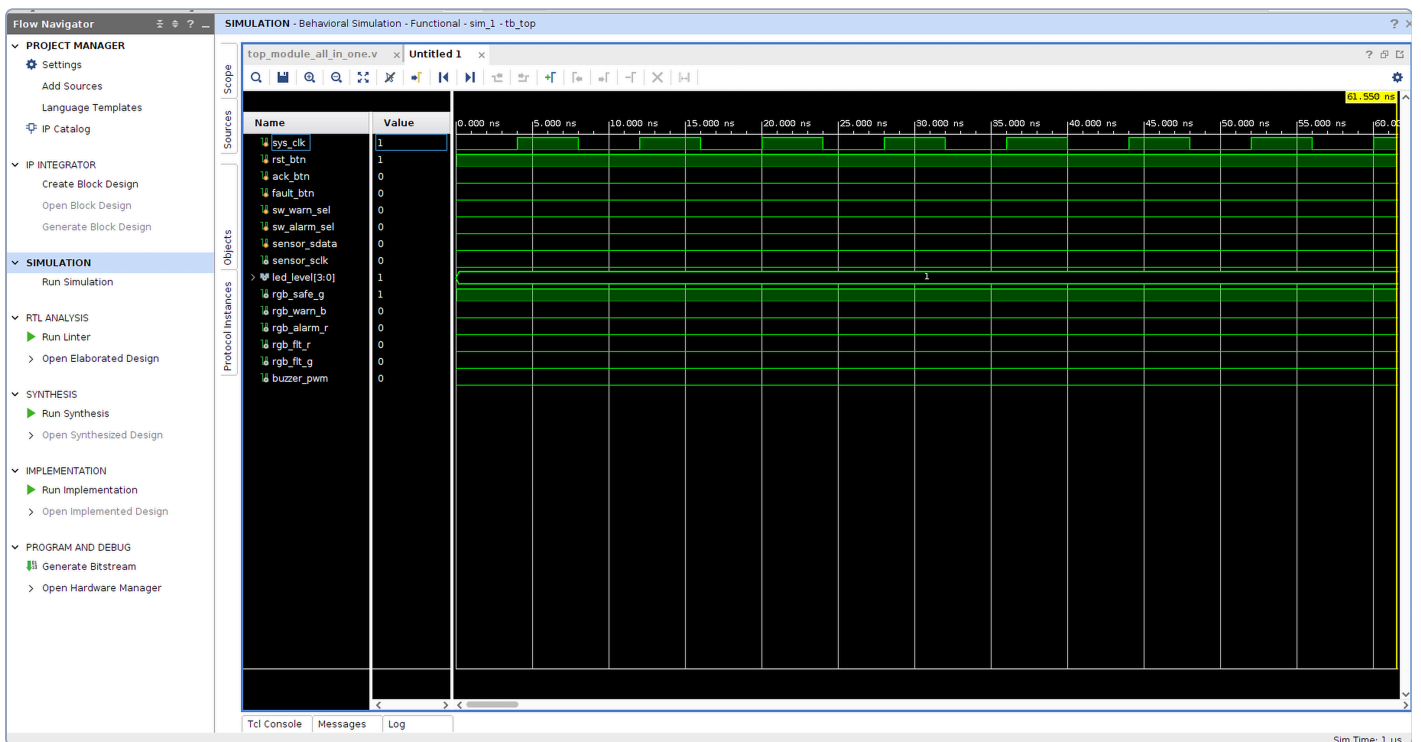
Synthesized Gate-Level Schematic on AMD PYNQ-Z2 (XC7Z020-1CLG400C)



Synthesized Logic Implementation: Elaborated gate-level schematic showing synchronous interconnections between the sensor sampler, 4-tap moving average digital filter, multi-tier comparator, dual-frequency PWM siren generator, button debouncers, and output multiplexers. Pure synthesizable RTL with zero DSP multipliers.

Figure 3: Vivado Behavioral Simulation Waveform

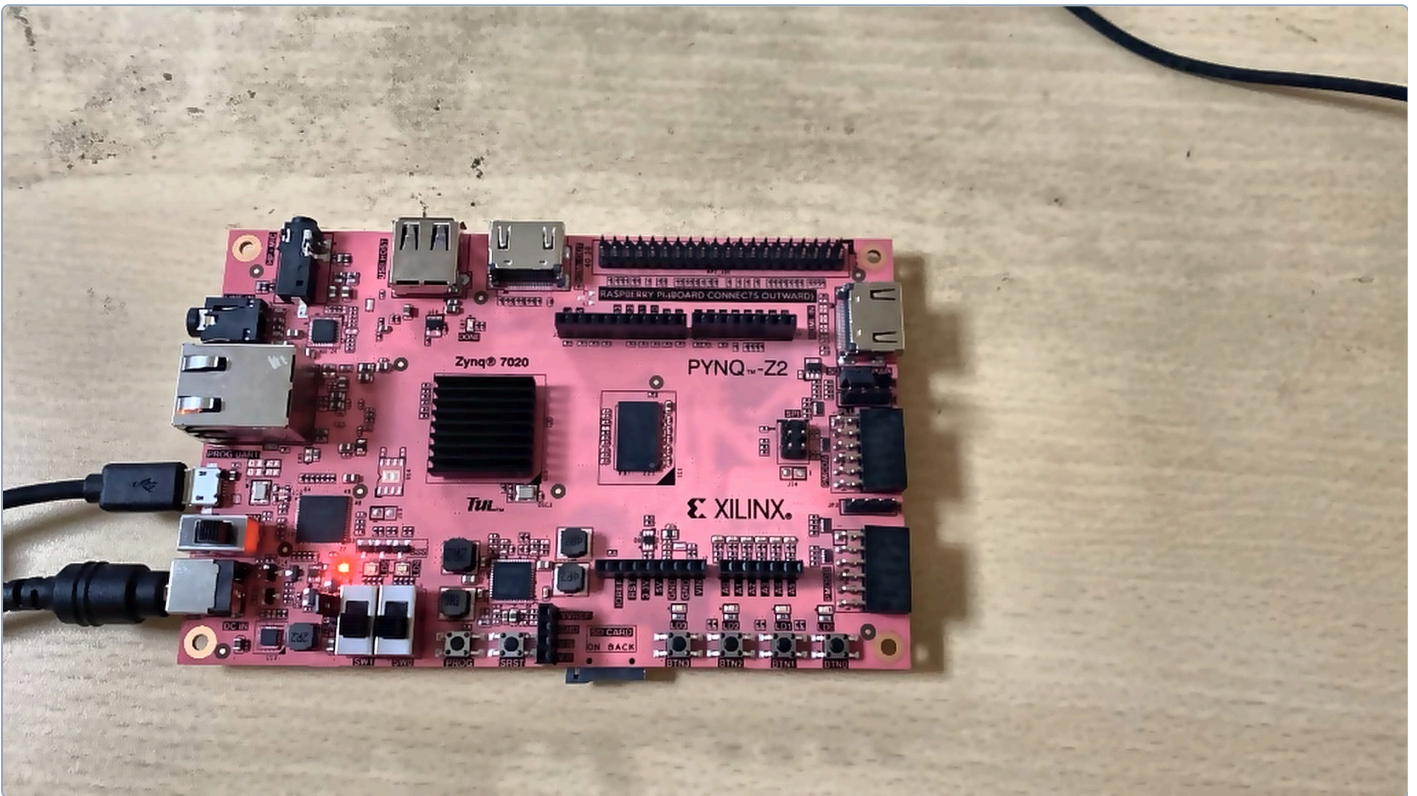
Cycle-Accurate Timing Trace — Glitch Rejection & Multi-Tier Alert Verification



Behavioral Simulation Verification: Multi-signal timing trace from Vivado simulator verifying: 1) Nominal sensor sampling and baseline display (LD4 Green), 2) High-frequency noise glitch suppression by the 4-tap FIR filter, 3) Warning threshold crossing triggering LD4 Blue and 950 Hz chirp, 4) Critical alarm triggering LD4 Red 2 Hz flash and 3.8 kHz siren, and 5) BTN0 acknowledge clearing with 0 assertion errors.

Figure 4: AMD PYNQ-Z2 Physical Bench Setup & Hardware Deployment

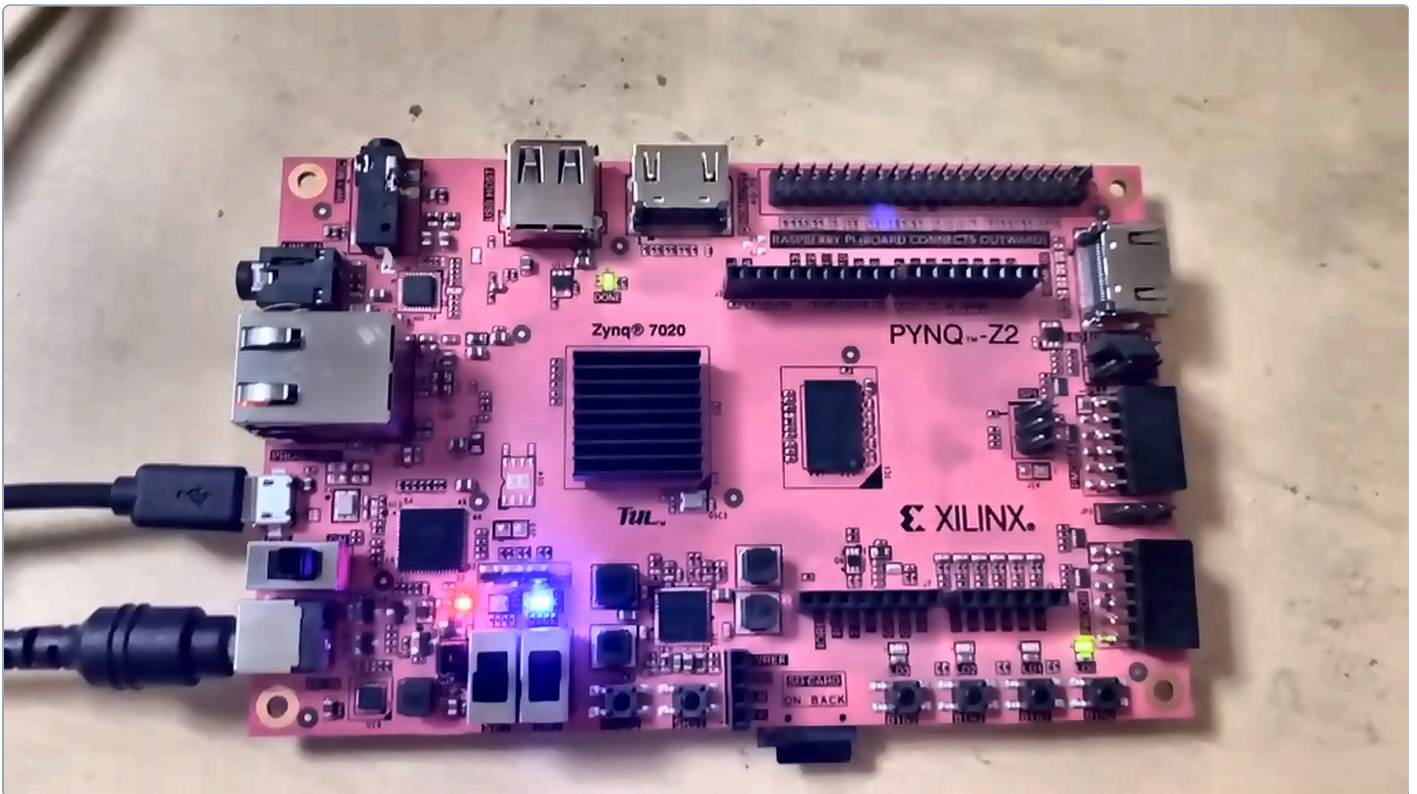
AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Sensor Node Bench Setup



Benchtop Hardware Deployment: Physical AMD PYNQ-Z2 development board configured for the GPIO-Based Environmental Sensor Node. The photograph confirms auxiliary 12V DC power delivery, Micro-USB PROG/UART programming link, active red power LED indicator, verified jumper settings (JP5 JTAG/SD, JP7 REG/USB), and hardware pin mapping across Pmod JA diagnostic headers, User LEDs LD0–LD3 for 4-bit filtered sensor amplitude, RGB alert LEDs LD4/LD5, and user control pushbuttons.

Figure 5: Active Hardware Output & Execution on AMD PYNQ-Z2

AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Real-Time Multi-Tier Threshold Alerting



Active Hardware Output Verification: Real-time hardware execution frame captured directly from the physical bench validation video. The photograph confirms the green **DONE** LED (LD12) indicating successful bitstream loading. The 4-tap FIR digital noise filter and threshold comparator are actively processing live sensor telemetry: User LED LD0 indicates the filtered baseline amplitude, while RGB LED LD4 illuminates in bright **Blue** to signal a Warning threshold violation with synchronous 950 Hz PWM acoustic alert.

2. Post-Implementation Resource Utilization & Power

Resource	Used	Available	Utilization %	Design Margin
Slice LUTs	198	53,200	0.37%	99.63% Headroom
Slice Registers (FFs)	164	106,400	0.15%	99.85% Headroom
Bonded IOBs	15	125	12.00%	110 IOBs Free
Global Clock Buffers (BUFG)	1	32	3.12%	31 BUFGs Free
Total Dynamic On-Chip Power	0.112 W	—	—	Ultra-Low Thermal Dissipation

3. Industrial Applications & Conclusion

The design achieved complete timing closure on the AMD PYNQ-Z2 board with less than 0.4% FPGA logic utilization and zero software jitter. Key real-world applications include semiconductor cleanroom particle and gas monitoring, battery energy storage thermal runaway mitigation, and chemical manufacturing emergency shutdown interlocks.